



Spectroscopic and laser properties of Tm:CNGG crystals grown by the micro-pulling-down method

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ABSTRACT

Tm³⁺-doped calcium niobium gallium garnet (Tm:CNGG) single crystals have been successfully grown by the micro-pulling-down (μ-PD) method. The crystal structure, absorption spectra, fluorescence spectra and lifetimes of Tm:CNGG crystals were investigated. The absorption and emission bandwidths of Tm:CNGG crystals are larger than those of other Tm³⁺ doped garnet crystals, which will be beneficial for laser diode (LD) pumping and ultrashort pulse generation. Additionally, continuous wave (CW) laser operation around 2 μm was achieved in the 2.0 at.% Tm:CNGG crystal by LD pumping.

1. Introduction

Lasers operating at 2 μm eye-safe wavelength region are widely applied to medicine, environmental detection, remote sensing, bioengineering, optical communication, and military affairs [1–4]. High-performance, low cost AlGaAs diode lasers operating near 800 nm arise the development of Tm³⁺ doped materials for all-solid-state lasers, where the Tm³⁺ was excited to ³H₄ energy level, followed by the cross relaxation process (³H₄→³F₄ and ³H₆→³F₄), that leads to the excitation of two ions to the upper level, for each pump photon [5]. Therefore, a large number of studies have been devoted to Tm³⁺ doped materials such as garnets [6,7], fluorides [8–11], sesquioxides [12] and tungstates [13,14]. Especially, sub-100-fs pulses have been realized in Tm:MgWO₄ (86 fs) crystal [15], Tm:(Lu_{2/3}Sc_{1/3})₂O₃ (63 fs) ceramic [16], Tm,Na:CNGG (84 fs) [17], and Tm:CLNGG (78 fs) crystals [18].

CNGG crystal was synthesized in 1985 for the first time [19] and its melting point is about 1460 °C. This makes it possible to synthesize these garnets using platinum crucibles, which simplifies the growth process and decreases the cost of the production in comparison with other garnet crystals. In addition, CNGG crystal is the representative disordered crystal due to the random distribution of Nb⁵⁺, Ga³⁺ and cationic vacancies in the host lattices. The disordered crystal lattice field in the host will cause significant inhomogeneous broadening in the

absorption and luminescence spectra of the rare-earth activator ions [20]. Broad absorption bands, resulting in a good matching of the absorption spectra of the crystals to the flashlamp and laser diodes emission, have made it possible to improve the pump efficiency of lasers and broad luminescence bands are favorable for the development of tunable and ultrashort lasers.

To date, the spectral and laser properties of Tm:CNGG disordered crystals have been studied [21–25]. The Tm:CNGG laser was reported with a low output power of 20 mW [21]. Most of Tm:CNGG crystals were grown by Czochralski (CZ) method. Unlike the conventional crystal growth methods, the μ-PD method is a versatile technique that allows crystal growth at much faster rate without cracks and much lower costs. Moreover, the shape of grown crystal is controllable, which simplifies the processing in comparison with bulk crystals. Therefore, it is very promising method for growing gain media for miniature lasers. In 2000, Tsuboi et al. grew Tm:CNGG single crystals by the μ-PD method and absorption, luminescence spectra and energy level were investigated carefully. Nonetheless, the laser performance has not been reported [22].

In this work, Tm:CNGG crystals were successfully grown by the μ-PD method. Room-temperature absorption spectra, emission spectra and lifetimes of Tm:CNGG crystals were measured and discussed. Furthermore, the laser operation of Tm:CNGG crystal grown by the μ-

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PD method has been investigated for the first time to the best of our knowledge.

2. Crystal growth and structure

Tm^{3+} -doped CNGG single crystals were grown by the μ -PD method. $\text{Ca}_3\text{Nb}_{1.6875}\text{Ga}_{3.1875}\text{O}_{12}$ was selected as the composition in order to keep charge balance close to its congruent melt composition. Raw materials of CaCO_3 , Nb_2O_5 and Ga_2O_3 with purity of 99.99% were weighed according to the ratio of 36.472:27.242:36.286 (wt.%) and Tm_2O_3 powders were doped together with additional Ga_2O_3 in a ratio of 3:5 (mol%) to form $\text{Tm}_3\text{Ga}_5\text{O}_{12}$. Tm-doping as a garnet phase is important to maintain garnet composition in melt. On the basis of spectral studies [22,25] and laser experiments previously performed on Tm:Na:CNGG [17], the Tm^{3+} concentrations of 2.0 at.% and 4.0 at.% have been selected in CNGG crystal. The mixture powders were compressed into bulks and loaded into an alumina container for sintering at 1300 °C for 12 h in a muffle furnace. The polycrystalline Tm:CNGG was melted within a platinum crucible, and then passed through the micro-nozzle at the bottom of the crucible to form a rod shaped micro single crystal. The outer diameter of crucible die was about 2 mm. Undoped $\langle 111 \rangle$ oriented CNGG crystal was used as the seed at the bottom of the crucible and the pulling rate was 0.5 mm/min. 2.0 at.% Tm:CNGG crystal with diameter of 2 mm and length of 68 mm and 4.0 at.% Tm:CNGG crystal with diameter of 2 mm and length of 59 mm were obtained as shown in Fig. 1 (a) and (b). The crystals were transparent with a light yellow color.

The structure of as-grown Tm:CNGG crystals was confirmed by X-ray diffraction (XRD) using an automated Ultima IV diffractometer (Cu target, $\text{K}\alpha$, Rigaku, Japan) at a scan width of 0.10° within $2\theta = 10\text{--}90^\circ$. Fine ground powders of the Tm:CNGG crystals were used for XRD measurement. Fig. 2 presents the X-ray powder diffraction pattern of Tm:CNGG crystals. The diffraction peak positions and relative intensity of Tm:CNGG crystals match well with CNGG crystal. The crystalline phase is the cubic phase (Ia3d) garnet-type structure, isostructural to $\text{Ca}_3(\text{NbGa})_2\text{Ga}_3\text{O}_{12}$ garnet (ICSD Card No.261821) [26].

3. Spectral properties

The crystal samples were cut from the as-grown crystals and two surfaces perpendicular to the $\langle 111 \rangle$ -growth axis were polished with the thickness of 1.06 mm. The measurements were conducted under the same conditions. The transmission spectra were measured by a spectrophotometer (Cary 5000, UV-VIS-NIR) in the wavelength region of 250–2250 nm. The fluorescence spectra and the decay curves excited with 683 nm, were measured by Edinburgh Instruments FLS920C spectrophotometers. The excitation source is an optical parametric oscillator laser with excitation pulse length of 5 ns. All the measurements were carried out at room temperature.

Fig. 3 (a) illustrates the transmission spectra of Tm:CNGG crystals in the wavelength region of 250–2250 nm. The 2.0 and 4.0 at.% Tm:CNGG crystals have an optical transmittance of $\sim 81\%$ in the range of 560–2250 nm, proving that the crystals grown by the μ -PD method have

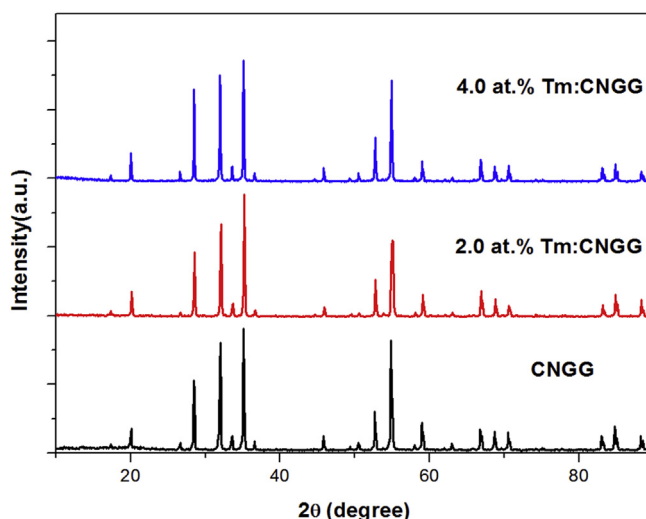


Fig. 2. Powder XRD pattern of Tm:CNGG crystals.

high optical quality. The absorption spectra of Tm:CNGG crystals are shown in Fig. 3 (b). The six main absorption bands of Tm^{3+} were found to be centered at around 358, 462, 683, 785, 1207 and 1640 nm, which correspond to the transitions from ground state $^3\text{H}_6$ to excited states $^1\text{D}_2$, $^1\text{G}_4$, $^3\text{F}_{2,3}$, $^3\text{H}_4$, $^3\text{H}_5$ and $^3\text{F}_4$, respectively. We observed no differences in the peaks shape and position between the samples grown by the μ -PD method and Czochralski method [25]. The absorption coefficients of 2.0 at.% Tm:CNGG crystal were calculated to be 2.3 cm^{-1} at 683 nm and 0.9 cm^{-1} at 785 nm, and those of 4.0 at.% Tm:CNGG crystal were 5.2 cm^{-1} at 683 nm and 1.8 cm^{-1} at 785 nm, respectively. The absorption coefficients increase with the increase of Tm^{3+} doping concentration. The full widths at half-maximum (FWHM) of the absorption peaks at 683 nm and 785 nm for two samples are similar, which are about 7 nm and 32 nm, respectively. The broad absorption band at 785 nm is much larger than that of Tm:YAG crystal (4 nm) [27] and Tm:YVO₄ crystal (5.0 nm) [28], which is suitable for LD pumping.

The fluorescence spectra of 2.0 and 4.0 at.% Tm:CNGG crystals in the range of 1500–2300 nm are presented in Fig. 4. The emission band corresponds to the transition from excited state $^3\text{F}_4$ to ground state $^3\text{H}_6$ of Tm^{3+} ions. It can be seen that the position of the strongest peak is very similar to that of the crystal grown by Czochralski method, but the peaks shape of fluorescence spectra in this work is weak due to the excitation source [25]. The FWHM of the emission band is about 329 nm, which is much larger than that of Tm:CaAlO₄ crystal (200 nm) [29], Tm:LiLuF₄ crystal (220 nm) [30] and Tm:LuYSiO₅ crystal (277 nm) [31]. The broad and smooth fluorescence spectra, arising from the disordered crystal structure, are beneficial for tunable and ultrafast laser operation around 2 μm wavelength.

The luminescence decay curves of $^3\text{F}_4$ level at 1860 nm excited by 683 nm are presented in Fig. 5. The decay curves of two crystals both exhibit a single exponential decay behavior. The decay times of 2.0 and 4.0 at.% Tm:CNGG crystals were fitted to be 4.35 ms and 1.46 ms. Time

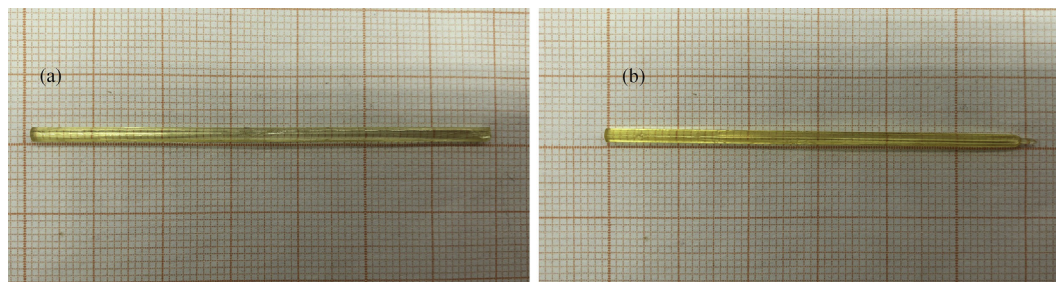


Fig. 1. Photographs of the as-grown 2.0 at.% Tm:CNGG (a) and 4.0 at.% Tm:CNGG (b) single crystals.

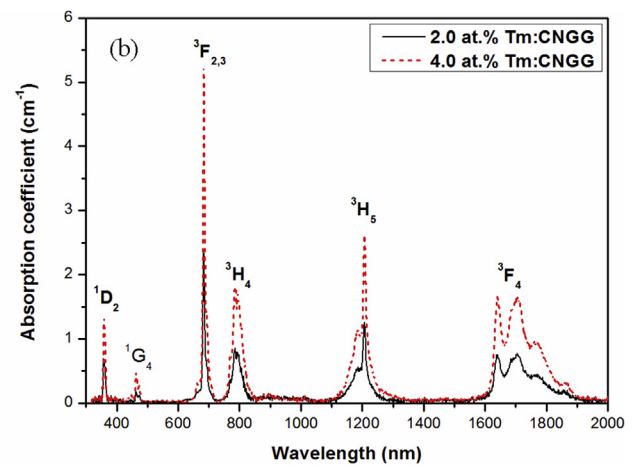
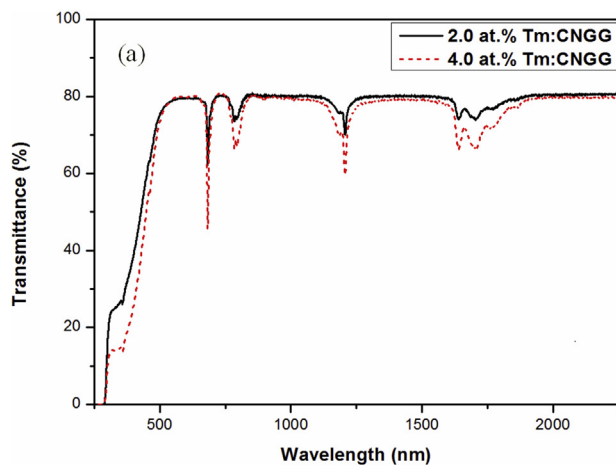


Fig. 3. Transmission spectra (a) and absorption spectra (b) of Tm:CNGG crystals.

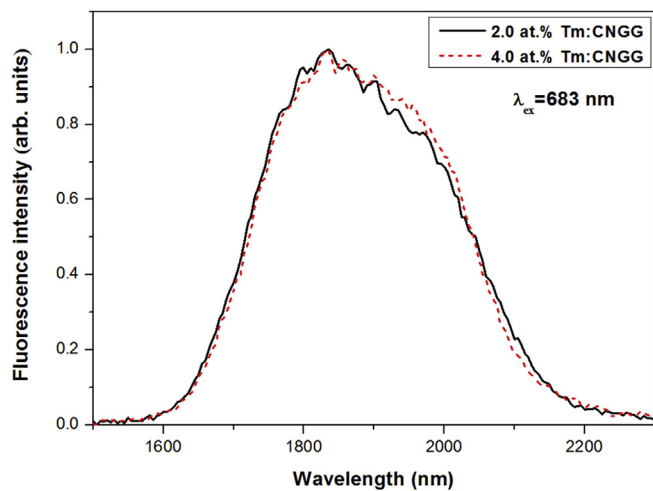


Fig. 4. Fluorescence spectra of Tm:CNGG crystals.

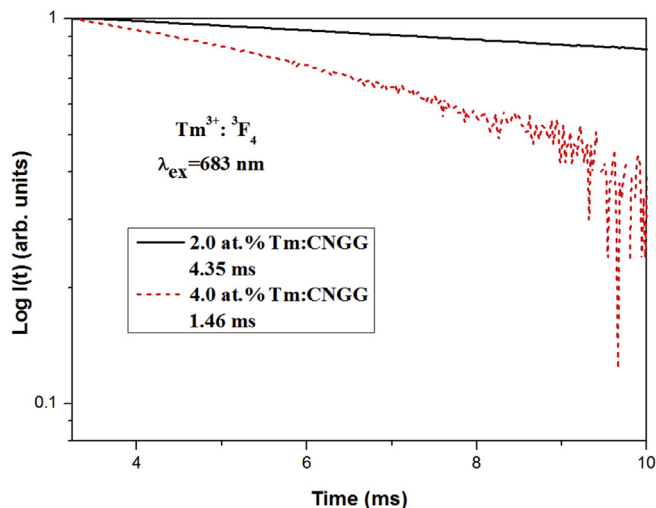


Fig. 5. The 1860 nm fluorescence decay curves of Tm:CNGG crystals.

during which the luminescence intensity decayed from level 3F_4 of Tm^{3+} ion decreases with the increases of Tm^{3+} doping concentration. The similar phenomenon has been reported in $Tm:LYSO_5$ crystal [31] and Tm^{3+} doped silicate glass [32]. The decay time values determined in our study are still larger than that of $Tm:YVO_4$ (0.8 ms) [28] and

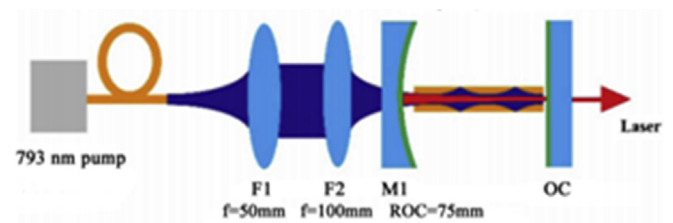


Fig. 6. Schematic of LD pumped Tm:CNGG crystal laser.

Tm^{3+} doped silicate glass (250 μs) [32], which proves that Tm:CNGG crystal is suitable for laser operation around 2.0 μm .

4. Laser performance

CW laser operation was carried out with the setup, as shown in Fig. 6. The pump source used in our experiment was a fiber-coupled LD at 793 nm, pigtailed to a 105/125 0.22 NA silica fiber. The pump beam was collimated and focused by a pair of lenses with focal length of 50 mm and 100 mm, respectively. The laser cavity consisted of a plane-concave mirror with a radius of curvature (ROC) of -75 mm and an output coupler (OC) with a transmission of 3%. A polished and uncoated 2.0 at.% Tm:CNGG crystal with diameter of 2 mm and length of 20 mm was used as the laser gain medium, which was wrapped with indium foil and tightly mounted in a water-cooled copper block. In the experiment, the cooling water temperature was sustained at 17 $^{\circ}C$.

Fig. 7 shows results of the CW output power as a function of the absorbed pump power. A maximum output power up to 174 mW was achieved with laser threshold of about 3.5 W of absorbed power, which leads to a slope efficiency of about 3.9%. Compared with Tm:CNGG bulk crystal [21], a higher output power was obtained in our experiment. The present experiment results could be improved by enhancing the crystal quality and optimizing the concentrations of Tm^{3+} and the cavity design in the future.

5. Conclusion

In summary, CNGG crystals with different Tm^{3+} concentrations have been successfully grown by the μ -PD method. The absorption coefficients of 2.0 and 4.0 at.% Tm:CNGG crystals at 785 nm are 0.9 cm^{-1} and 1.8 cm^{-1} , respectively. The FWHM of the absorption peaks at 785 nm of two crystals are both 32 nm, which are suitable for LD pumping. The FWHM of the emission band are about 329 nm and the fluorescence lifetimes of two crystals are 4.35 ms and 1.46 ms, respectively. CW laser operation around 2 μm was demonstrated in the 2.0 at.% Tm:CNGG crystal under LD pumping. The maximum output

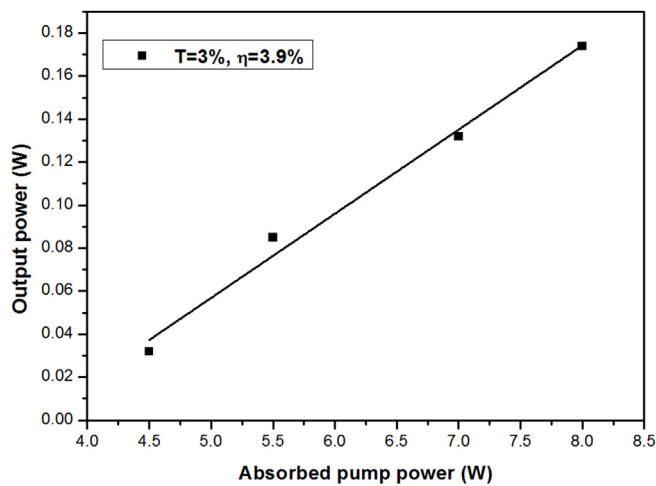


Fig. 7. Output power versus absorbed pump power with $T = 3\%$ output coupler transmission.

power of 174 mW was obtained, corresponding to the slope efficiency of 3.9%. The disordered Tm:CNCG crystal has wide absorption and emission bands, implying its potential for generation of ultrashort pulse at 2 μm eye-safe wavelength.

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